

Financial Analysis of GDP, VTI, VXX and Cryptocurrencies

Financial Analysis

Economic Factors Dashboard	Covid-19 Pandemic Trend	Economic Factors Data Limitations	Bitcoin Performance	BTC, VTI & VXX Comparison	Cryptocurrency Data Limitations
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How have GDP, VTI, and VXX Changed Over Time?

GDP and VTI generally moved upward over time, while VXX moved in the opposite direction — suggesting that economic growth and market confidence tend to rise when volatility expectations fall.

VTI
Total Stock Market ETF

211.7%

GDP Real
Economic Output

74.8%

VXX
Volatility ETN

-71.3%

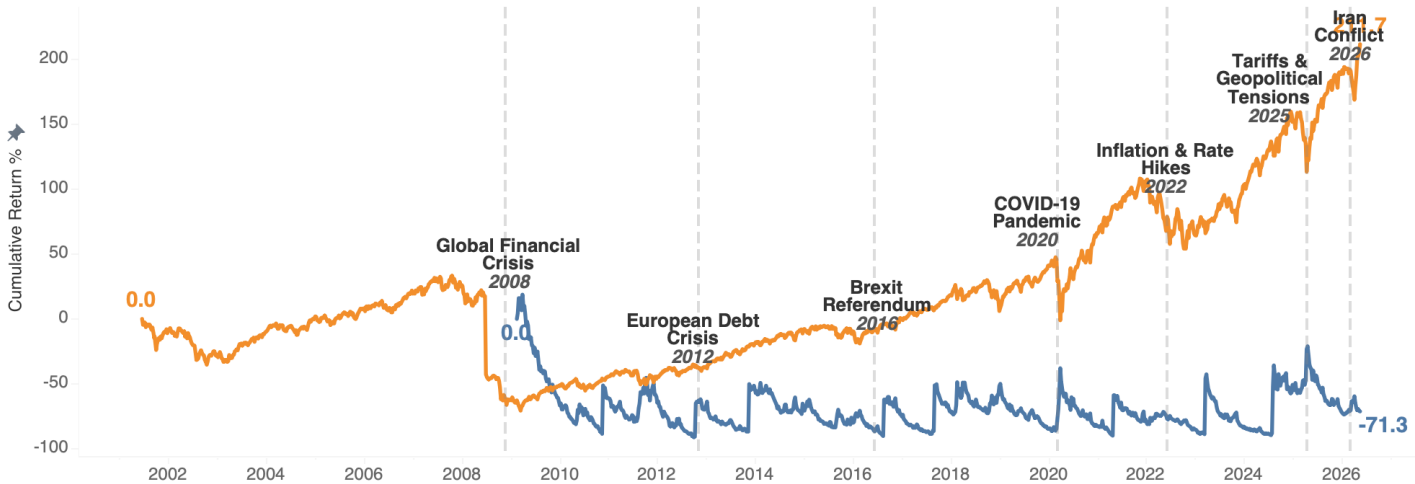
Investor Confidence Rose

Volatility Fell

Inverse Relationship Observed Since 2009

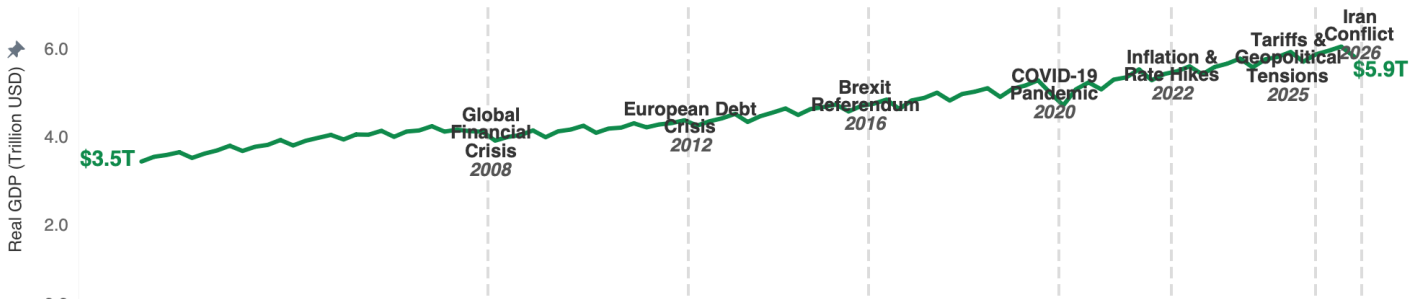
VTI Rose While VXX Fell: Market Confidence and Volatility Moved in Opposite Directions

Cumulative return highlights how VTI gained strongly over time while VXX declined overall, suggesting that stock market performance and volatility expectations often move counter to each other.



U.S. Economic Output Continued Rising Despite Major Economic Shocks

Real GDP increased from roughly \$3.5T in 2002 to \$6T by 2026, with visible slowdowns during the Global Financial Crisis and COVID-19 pandemic.



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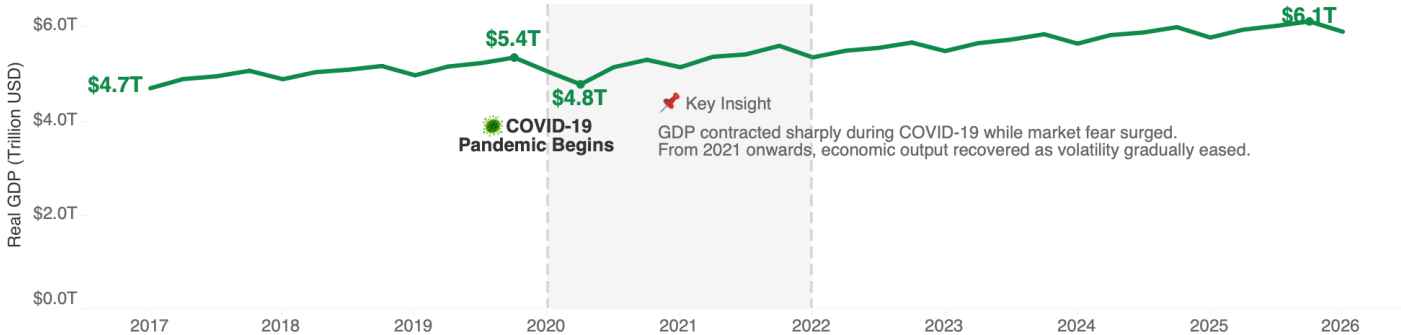
The Pandemic Disrupted Economic Growth, But Recovery Followed as Volatility Declined

Real GDP reflects economic output while VXX acts as a proxy for market fear and consumer uncertainty during and after the pandemic



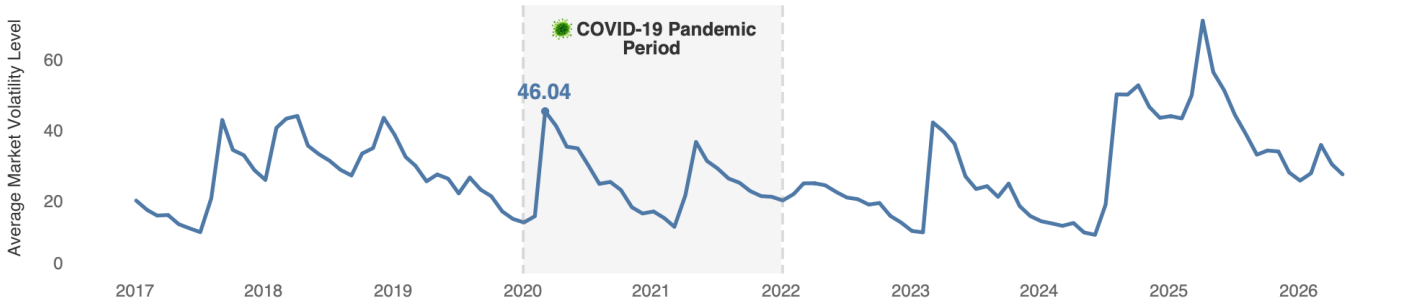
GDP Fell During the Pandemic Shock, Then Recovered

Real GDP was rising steadily before 2020, declined sharply during COVID-19, then resumed growth from 2021 onwards



VXX Spiked During COVID-19 as Market Fear Rose

VXX rose sharply during the pandemic shock, reflecting heightened volatility expectations, then declined as markets stabilized.



Key Takeaway: Before 2020, GDP followed a steady upward trend while VXX remained relatively stable. During the pandemic, GDP declined sharply as volatility surged, reflecting heightened uncertainty. From 2021 onwards, GDP recovered steadily while VXX moderated, suggesting improving economic conditions and stabilizing market sentiment.

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DATA LIMITATIONS

01 LIMITED INDICATORS

GDP, VTI, VXX, and consumer sentiment **do not capture unemployment, inflation, wage inequality, household debt, and small business health.**

02 DIFFERENT FREQUENCIES

- GDP: Quarterly
- VTI & VXX: Weekly
- BTC: Daily

This makes **direct comparison imperfect.**

03 MARKET $\neq$ ECONOMY

VTI reflects investor sentiment and large corporations. But stock market growth **may not reflect everyday household conditions, income inequality, and cost of living.**

04 VXX IS SHORT-TERM FEAR

VXX measures expected volatility, **not long-term economic health.**

05 GDP HAS LAG

GDP is **backward-looking, revised over time,** and **misses informal economic activity**

CONCLUSION

While these indicators provide useful signals about economic growth, investor confidence, and market volatility, they offer only a partial view of national economic health. A more complete assessment would require additional indicators such as employment, inflation, wages, household affordability, and business activity.

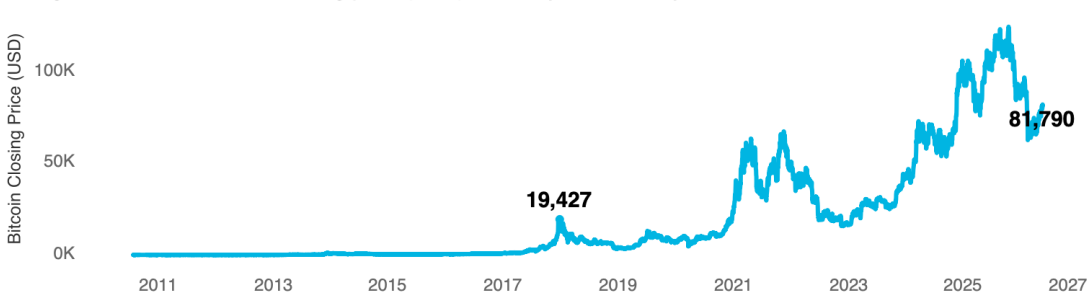
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A long-term view of Bitcoin's closing price (USD) from July 2010 to May 2026

Bitcoin Experienced Sharp Growth, Deep Pullbacks, and Renewed Recovery

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💡 Key Takeaways:

Extraordinary Long-Term Growth

Bitcoin has delivered -161 million percent in cumulative returns since July 2010, far outperforming traditional assets

Extremely Volatile Along the Way

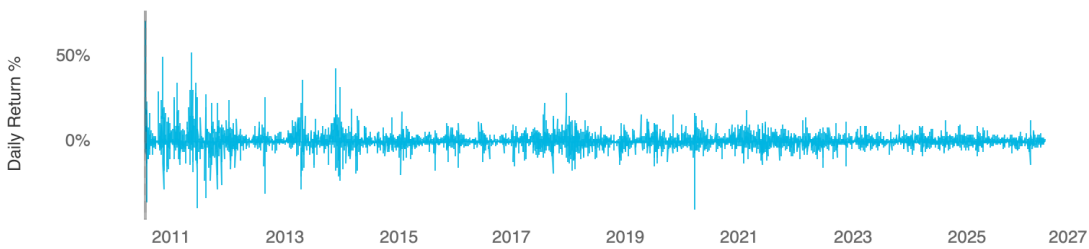
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Major Cycles, Stronger Over Time

Bitcoin has experienced multiple boom-and-bust cycles, with each recovery reaching new all-time highs.

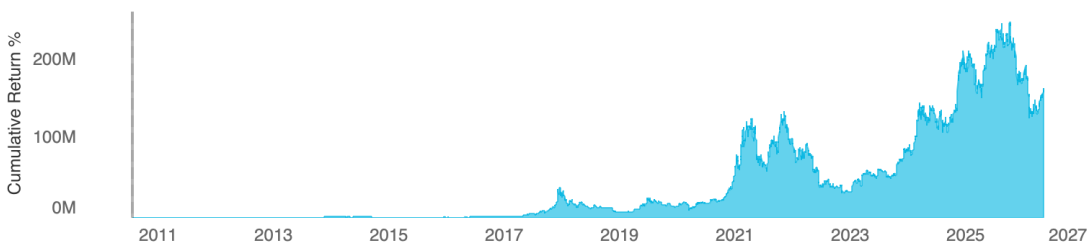
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Daily return percentage of Bitcoin (BTC) from 2010–2025



Bitcoin Delivered Strong Long-Term Gains Despite Volatile Swings

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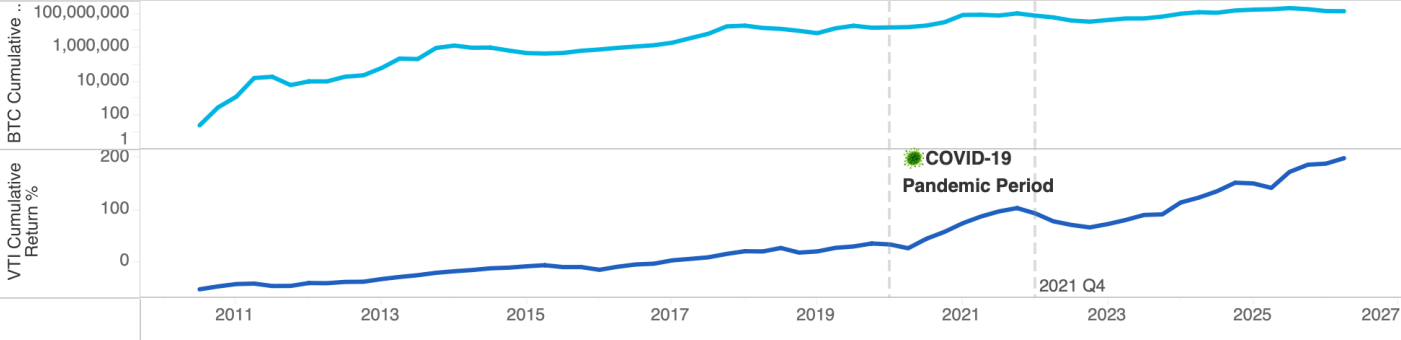
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Comparing Bitcoin's price movement against VTI and VXX over the last few years

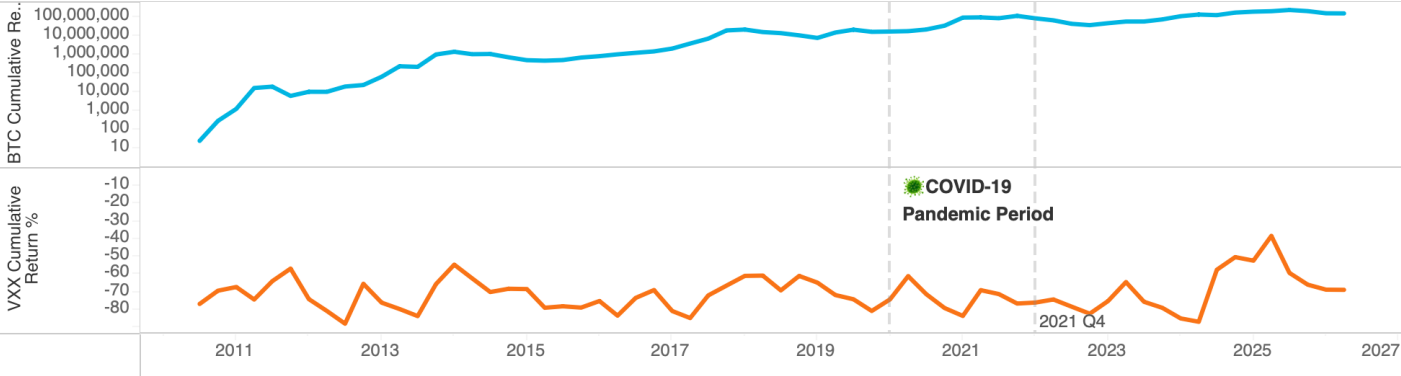
Bitcoin Outpaced VTI but with Much Larger Swings

Bitcoin generally moved with greater volatility than VTI. While VTI reflects broader stock market performance, BTC showed sharper rallies and deeper corrections.



Bitcoin Reacted Differently from VXX, the Market Volatility Indicator

VXX tends to rise during periods of market stress, while Bitcoin can behave more like a speculative risk asset. During some crisis periods, Bitcoin may fall while VXX spikes.



Key Observation: Bitcoin generally moved in the same long-term direction as VTI, reflecting broader growth in risk assets, but with significantly greater volatility and larger price swings. In contrast, Bitcoin showed an inverse or unstable relationship with VXX, often declining during periods when market volatility spiked, suggesting that Bitcoin behaves more like a speculative asset than a traditional safe-haven investment.

Note: BTC cumulative return uses a logarithmic scale to improve readability due to extreme long-term growth.

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DATA LIMITATIONS

01 ONLY BITCOIN IS ANALYZED

The dataset focuses **only on Bitcoin** and does not represent the behavior of the broader cryptocurrency market, including altcoins and stablecoins.

02 LACK OF MARKET CONTEXT

The data contains **only historical price and volume information** and does not include macroeconomic events, regulations, news, or investor sentiment that influence cryptocurrency prices.

03 HIGH MARKET VOLATILITY

Cryptocurrency markets are **highly volatile**, making historical patterns and trends less reliable for predicting future market movements.

04 DAILY-LEVEL GRANULARITY

The dataset uses **daily data**, which may hide important intraday price swings, flash crashes, and short-term trading activity.

05 INCOMPLETE VOLUME REPRESENTATION

Trading volume from a single API source **may not fully capture activity across all global cryptocurrency exchanges and decentralized platforms**.

CONCLUSION

While the dataset provides useful insights into Bitcoin's price trends, returns, and volatility, it does not fully represent the complexity of the cryptocurrency market. Important factors such as market sentiment, macroeconomic events, and broader crypto ecosystem behavior are not captured in the data. Therefore, conclusions drawn from **this dataset** should be interpreted with caution and complemented with additional sources of information.



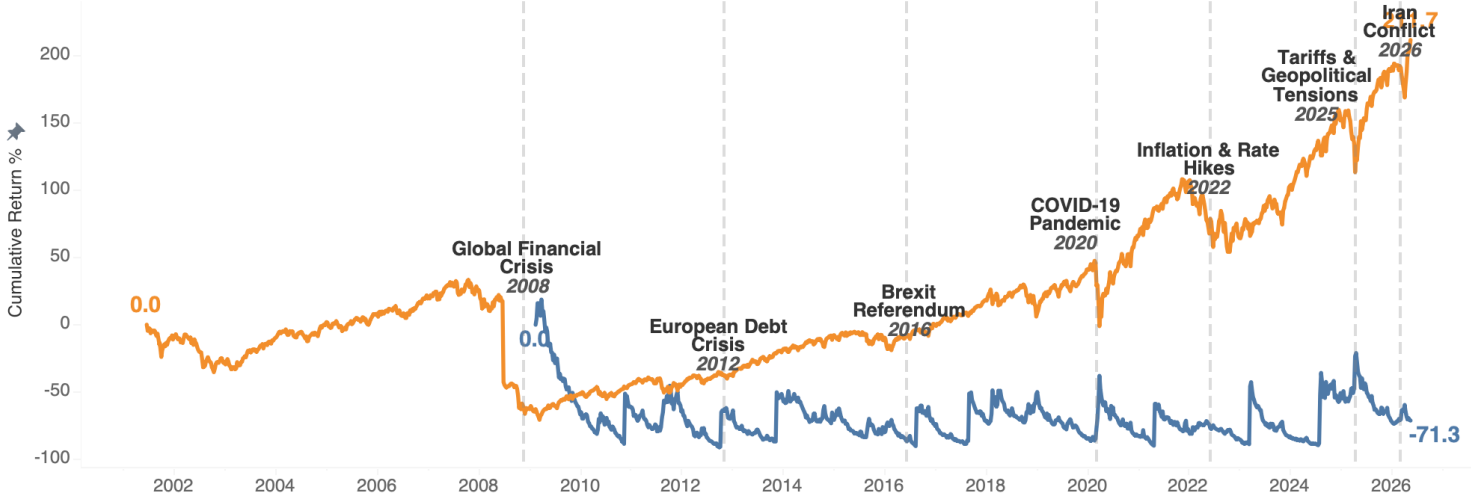
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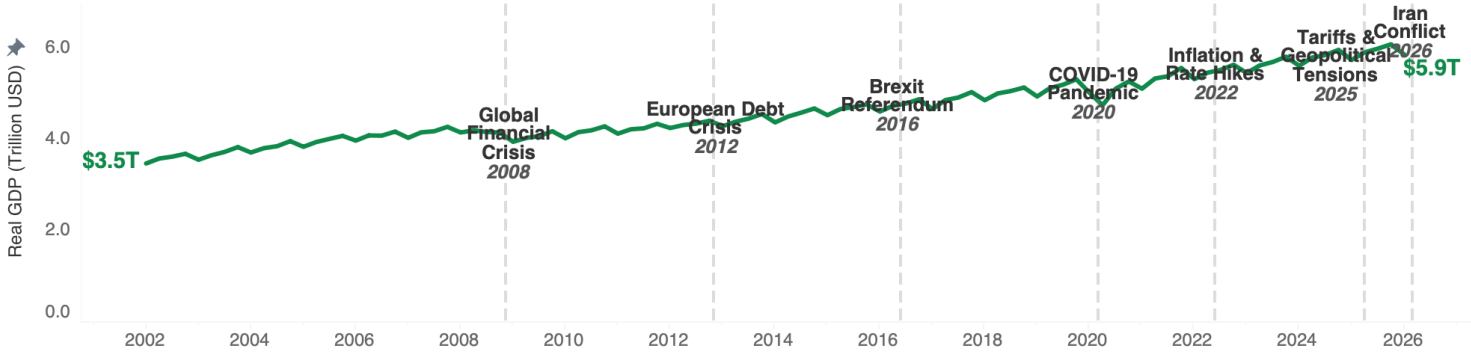
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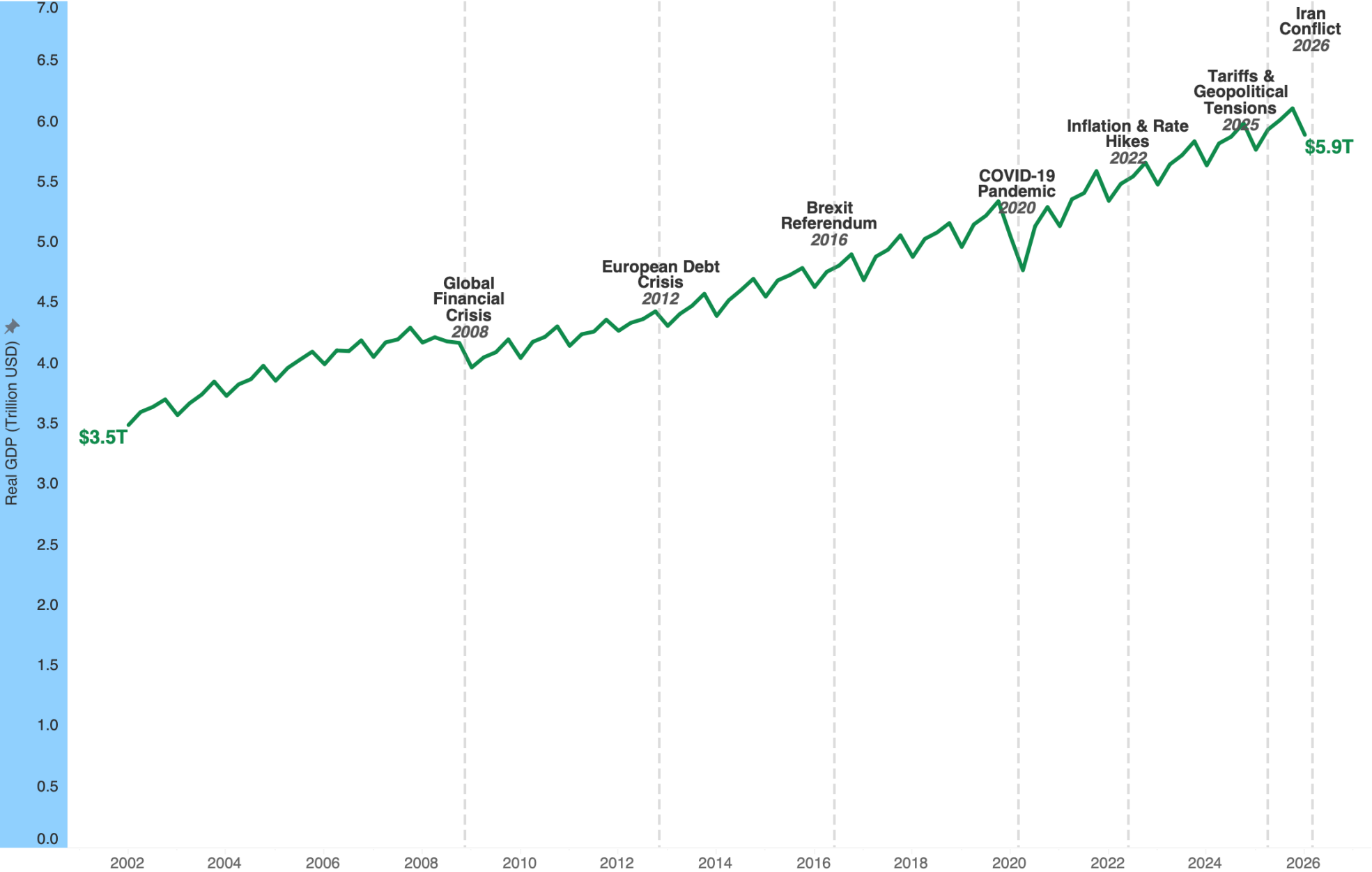
Real GDP increased from roughly \$3.5T in 2002 to \$6T by 2026, with visible slowdowns during the Global Financial Crisis and COVID-19 pandemic.



Key Takeaway: VTI and GDP broadly reflected long-term economic and market growth, while VXX moved counter to them as volatility eased over time. Together, these indicators suggest that investor confidence strengthened as economic conditions improved.

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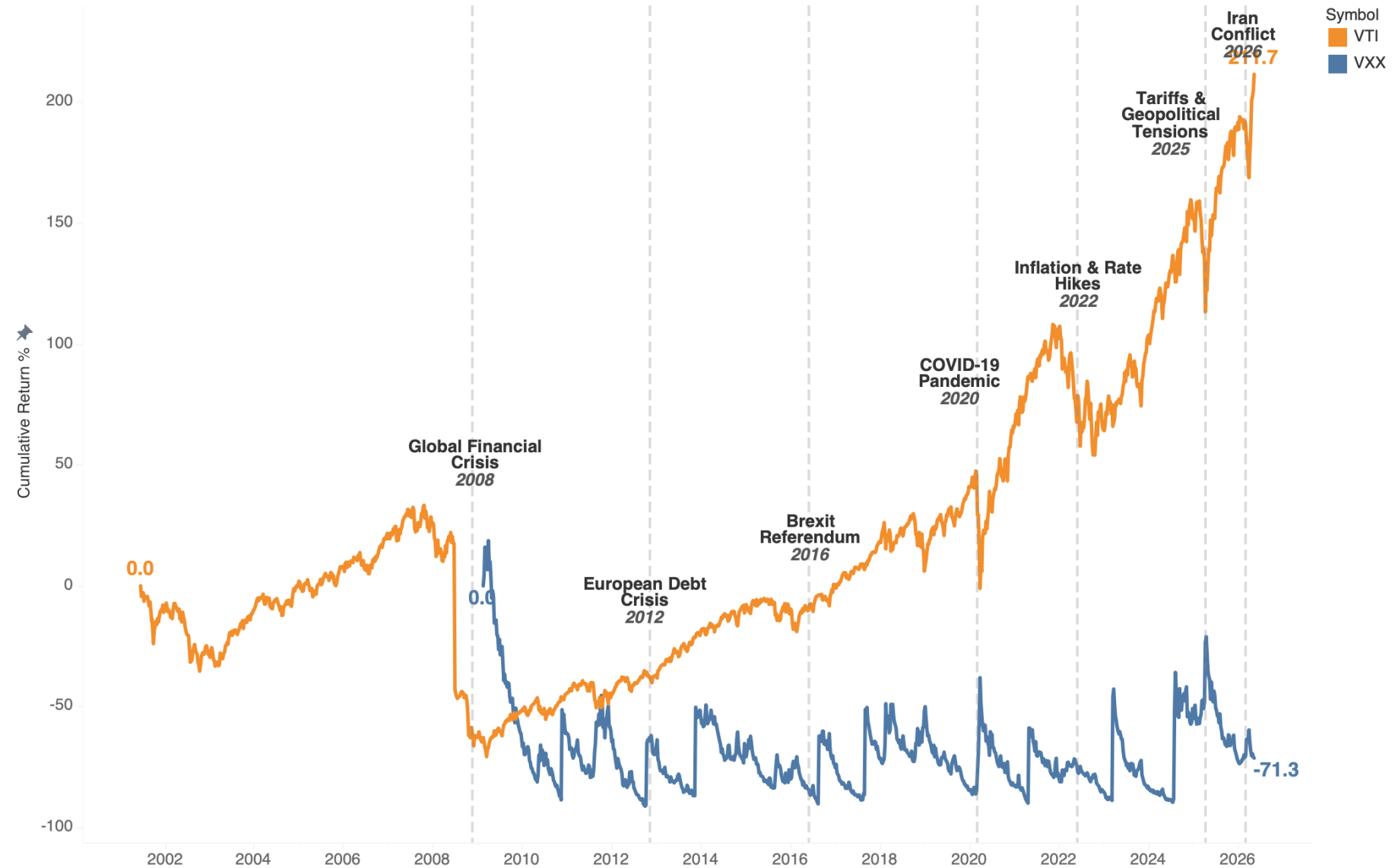
Real GDP increased from roughly \$3.5T in 2002 to \$6T by 2026, with visible slowdowns during the Global Financial Crisis and COVID-19 pandemic.



What this shows: Real GDP generally trended upward over the long term, reflecting continued expansion in economic output. However, GDP reacted more slowly to economic shocks because it is measured quarterly and reflects broader economic activity rather than short-term market sentiment.

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Cumulative return highlights how VTI gained strongly over time while VXX declined overall, suggesting that stock market performance and volatility expectations often move counter to each other.



What this shows:

VTI delivered strong long-term gains while VXX trended lower overall with periodic spikes during uncertainty. This inverse relationship suggests that investor confidence tends to rise when volatility expectations fall.

Why is Cumulative Return % better than Close?

Close price shows raw price movement, but VTI and VXX have very different price ranges and structures. Cumulative Return % shows performance relative to the starting point, making it much easier to answer: "Which went up, which went down, and are they moving together or counter to each other?"

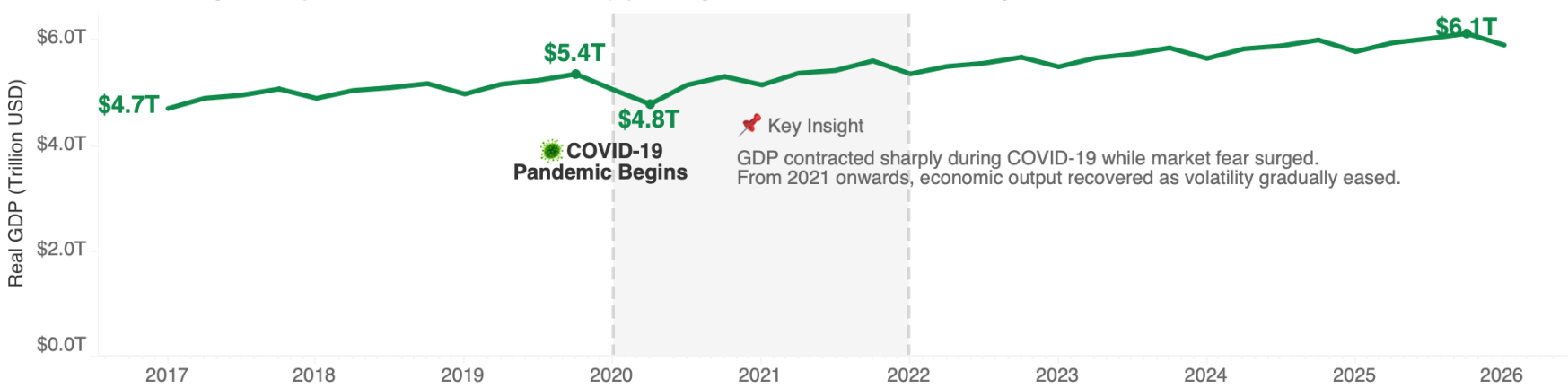
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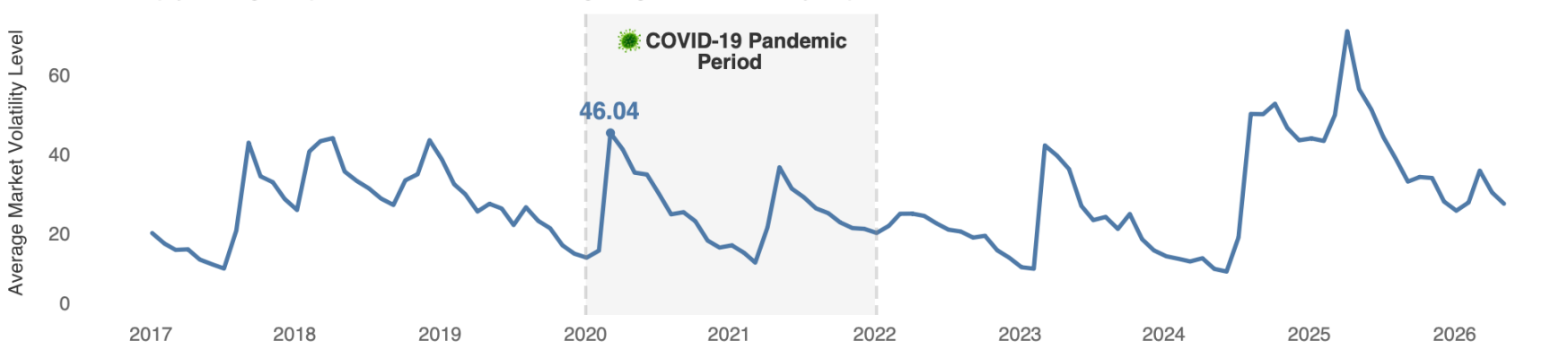
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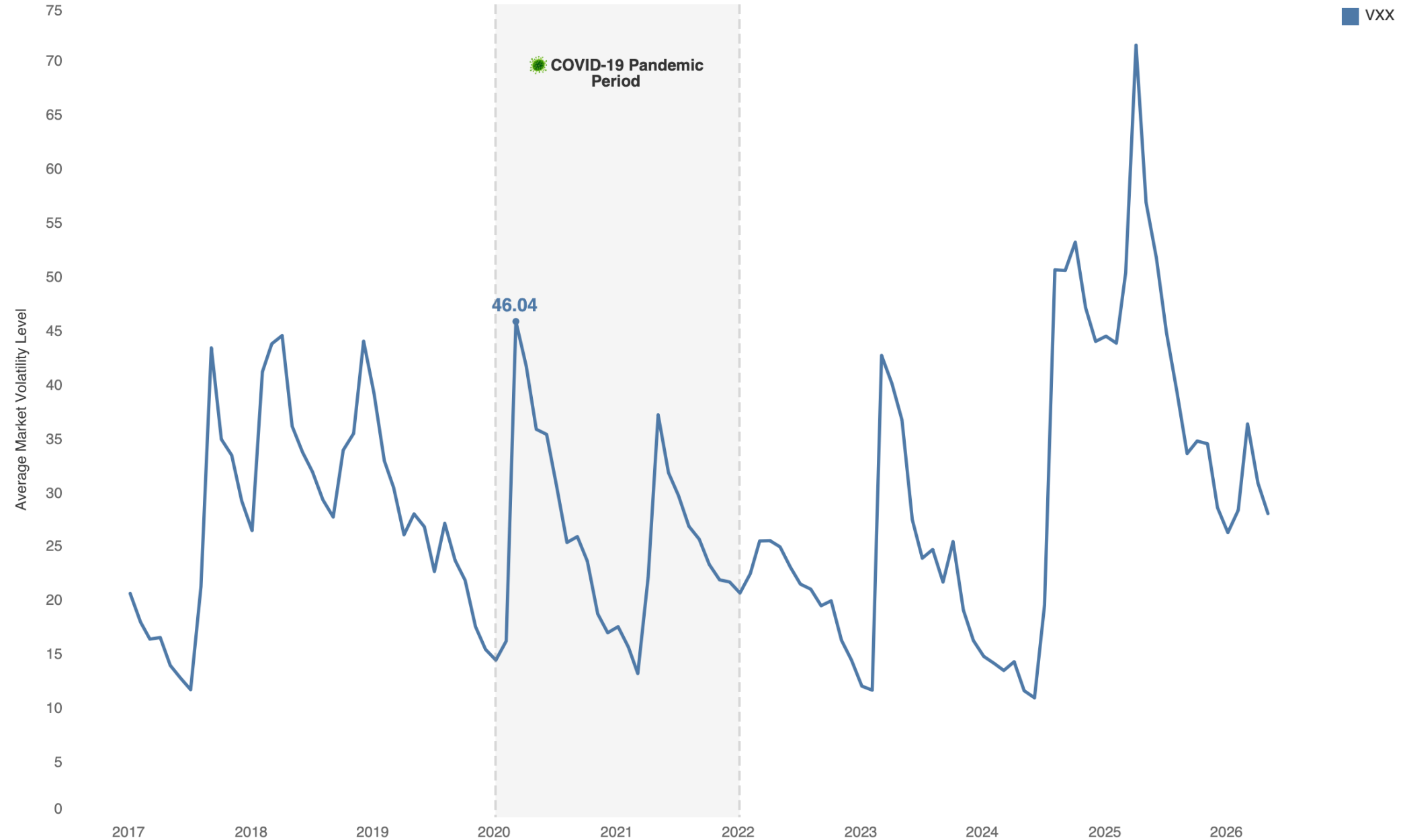
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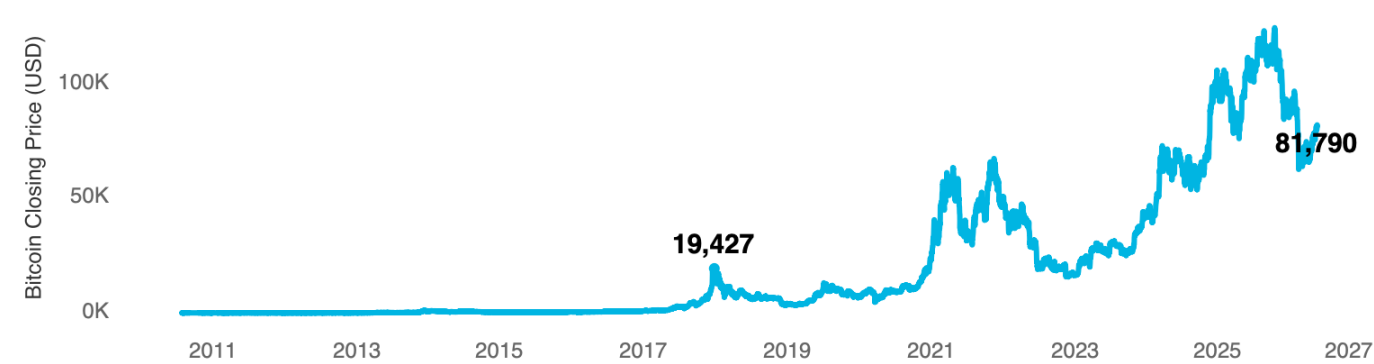
Before 2020, VXX was relatively contained with short-term spikes. During the COVID-19 shock, VXX surged as uncertainty increased. From 2020 onwards, VXX generally trended downward, suggesting volatility expectations eased over time.

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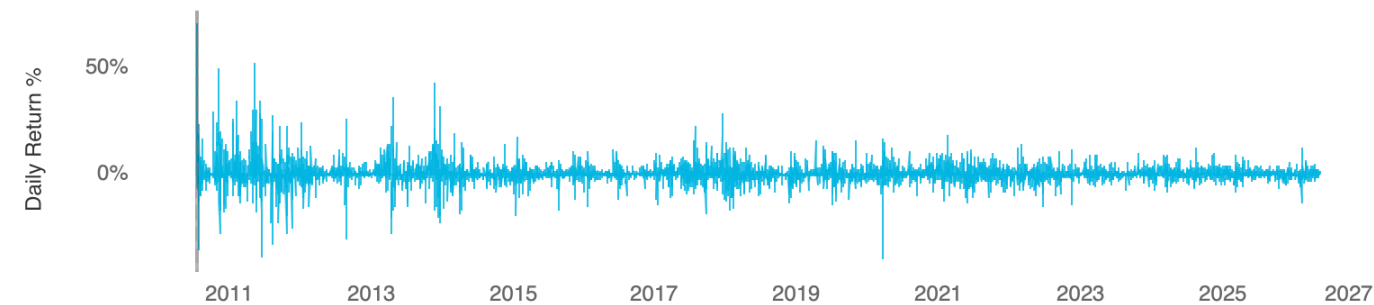
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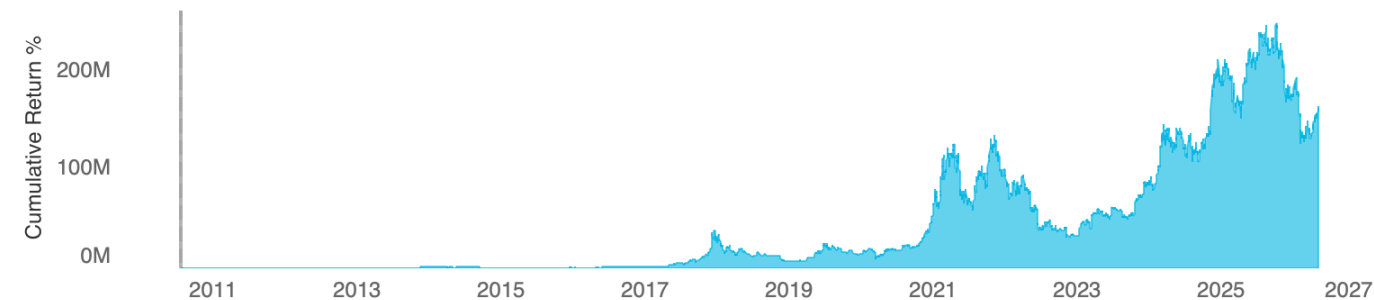
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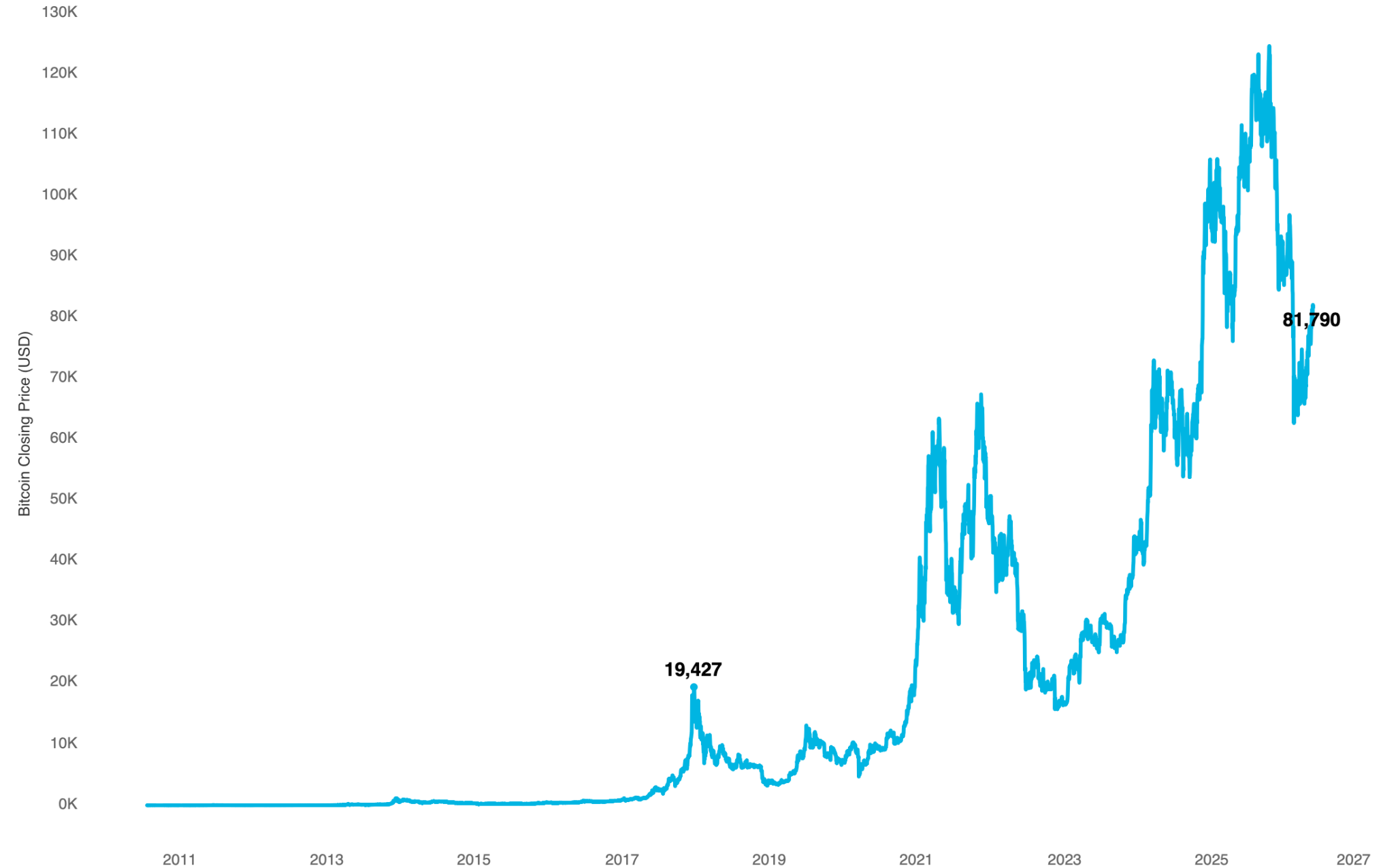
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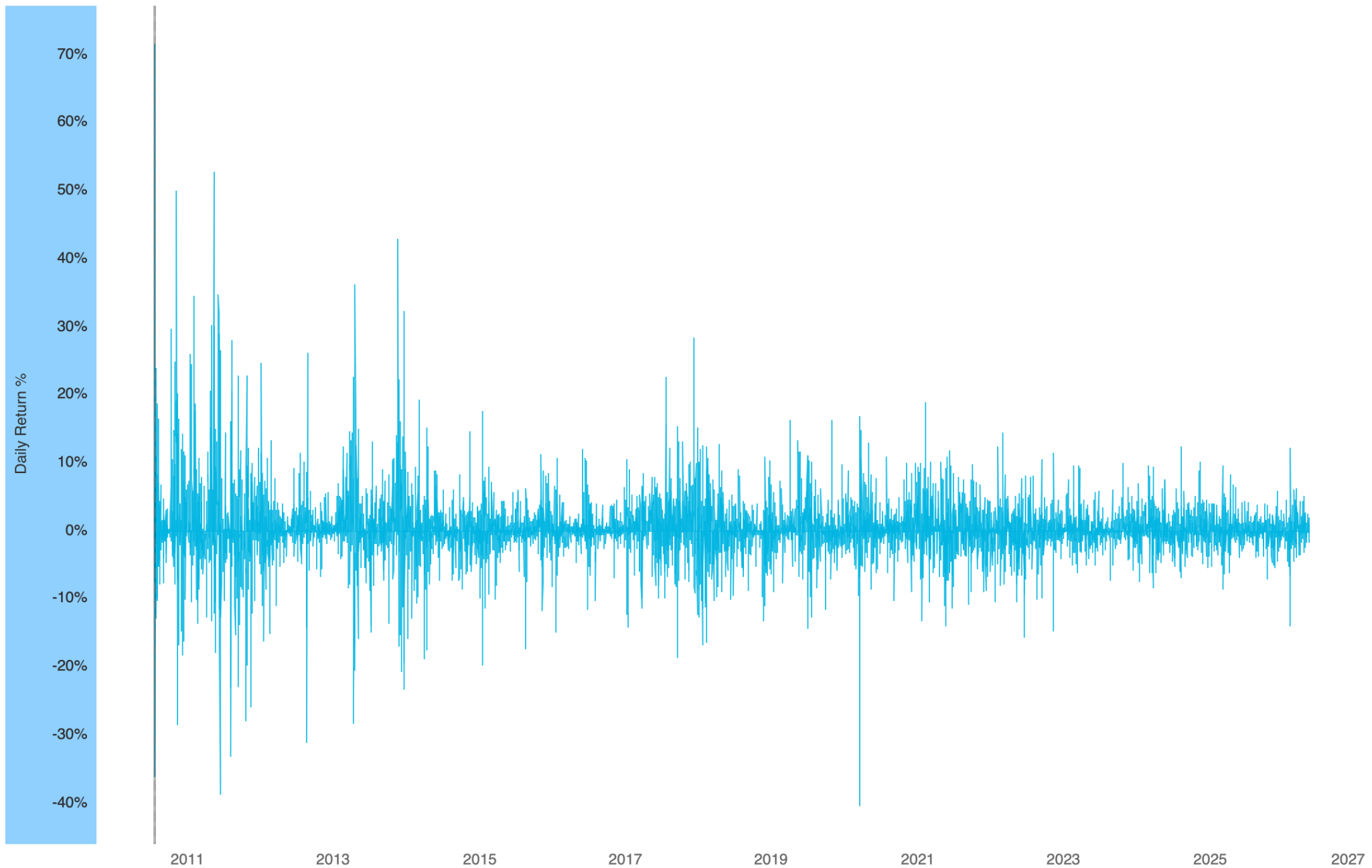
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Bitcoin has delivered extraordinary long-term growth, rising from near zero to over \$80K. However, this growth has come with significant volatility, including steep drawdowns during periods of market stress.

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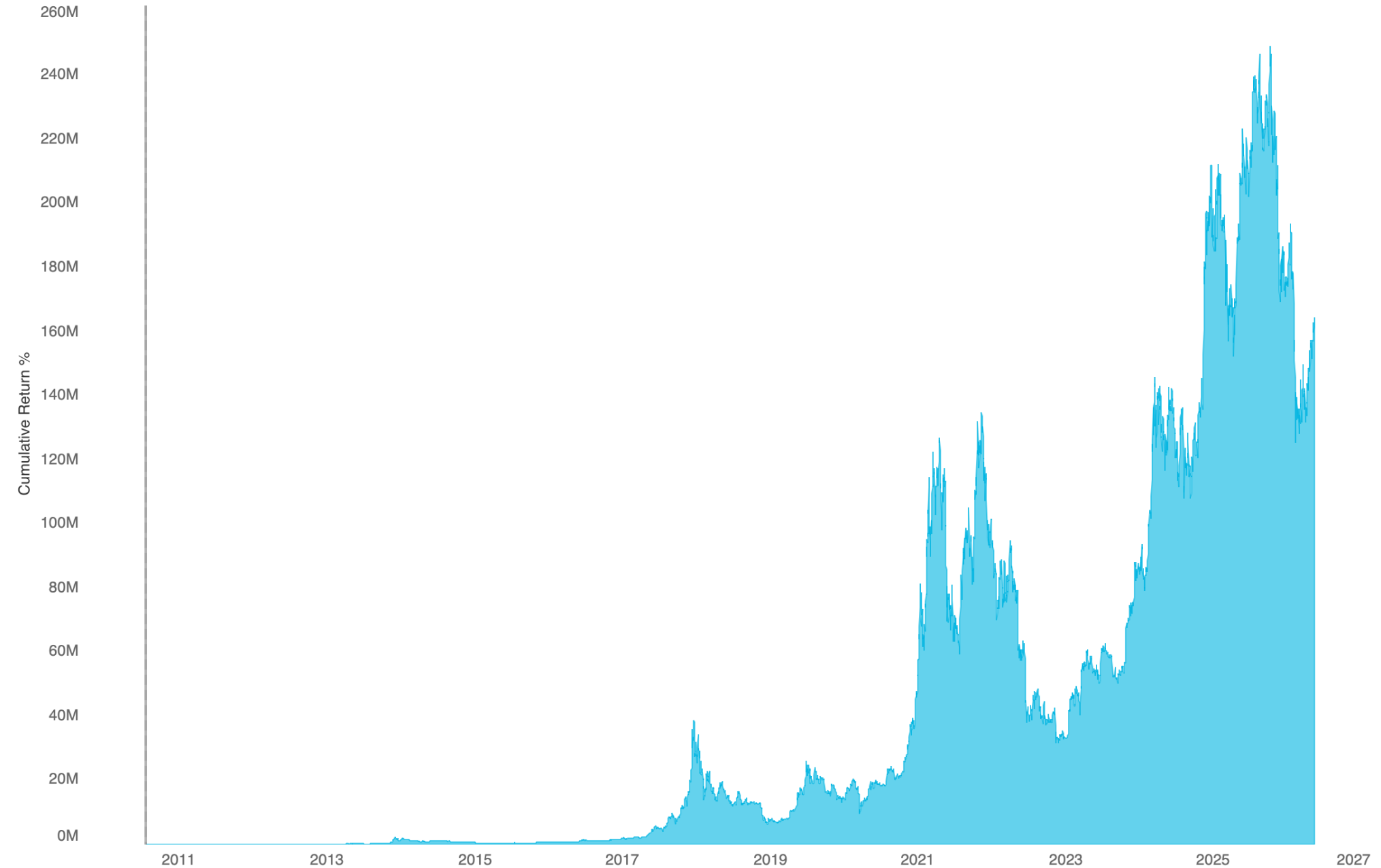
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Bitcoin experiences significantly larger daily price swings than traditional financial assets, reflecting its speculative and high-risk nature.

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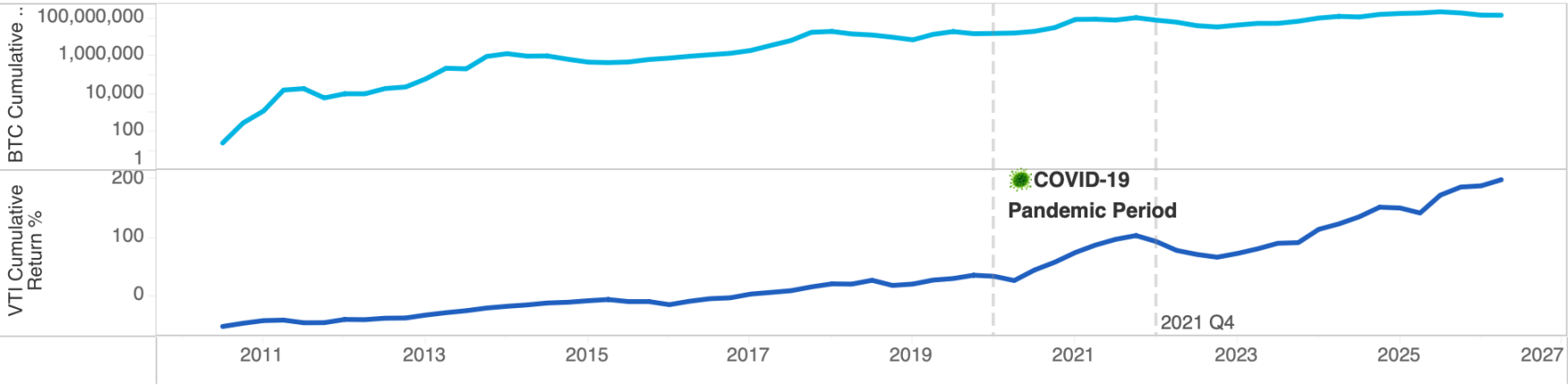
Bitcoin has delivered extraordinary long-term returns, rising over 161 million percent since 2010. However, this growth has come with extreme volatility and multiple drawdowns, highlighting the high-risk, high-reward nature of Bitcoin as an asset.

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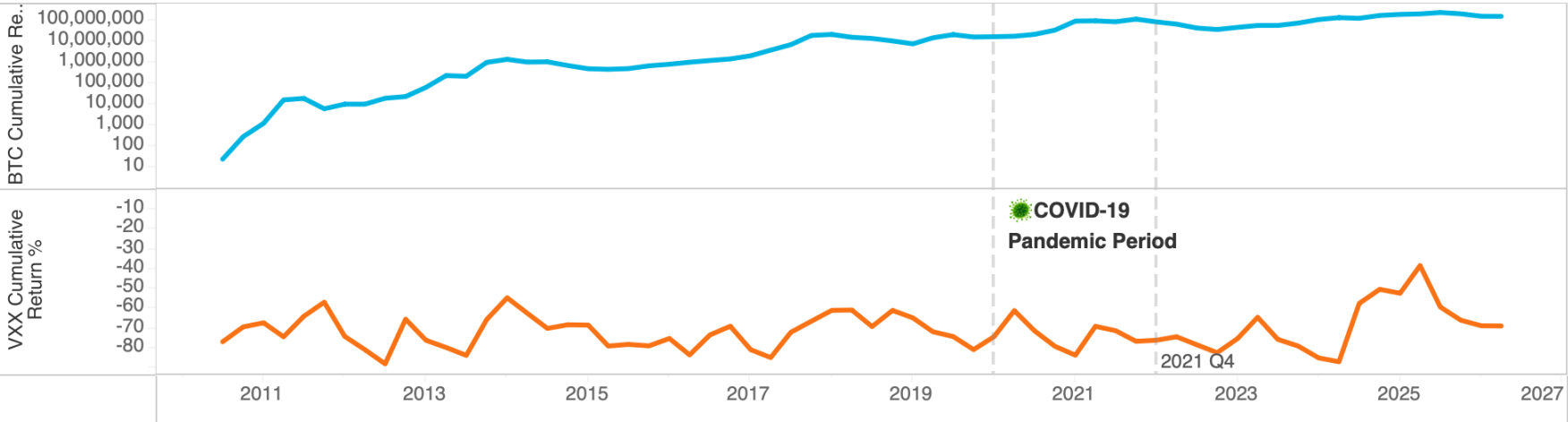
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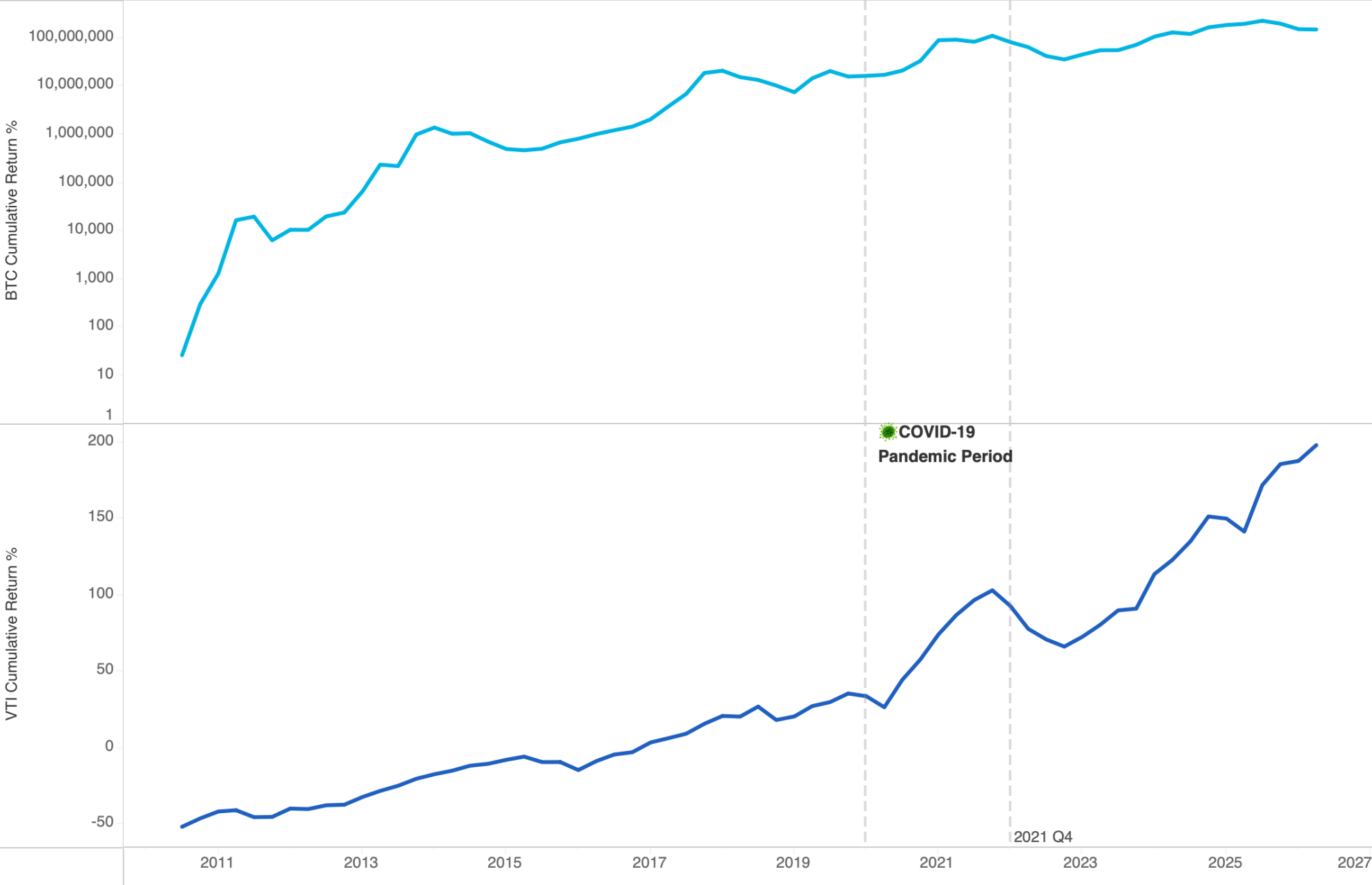


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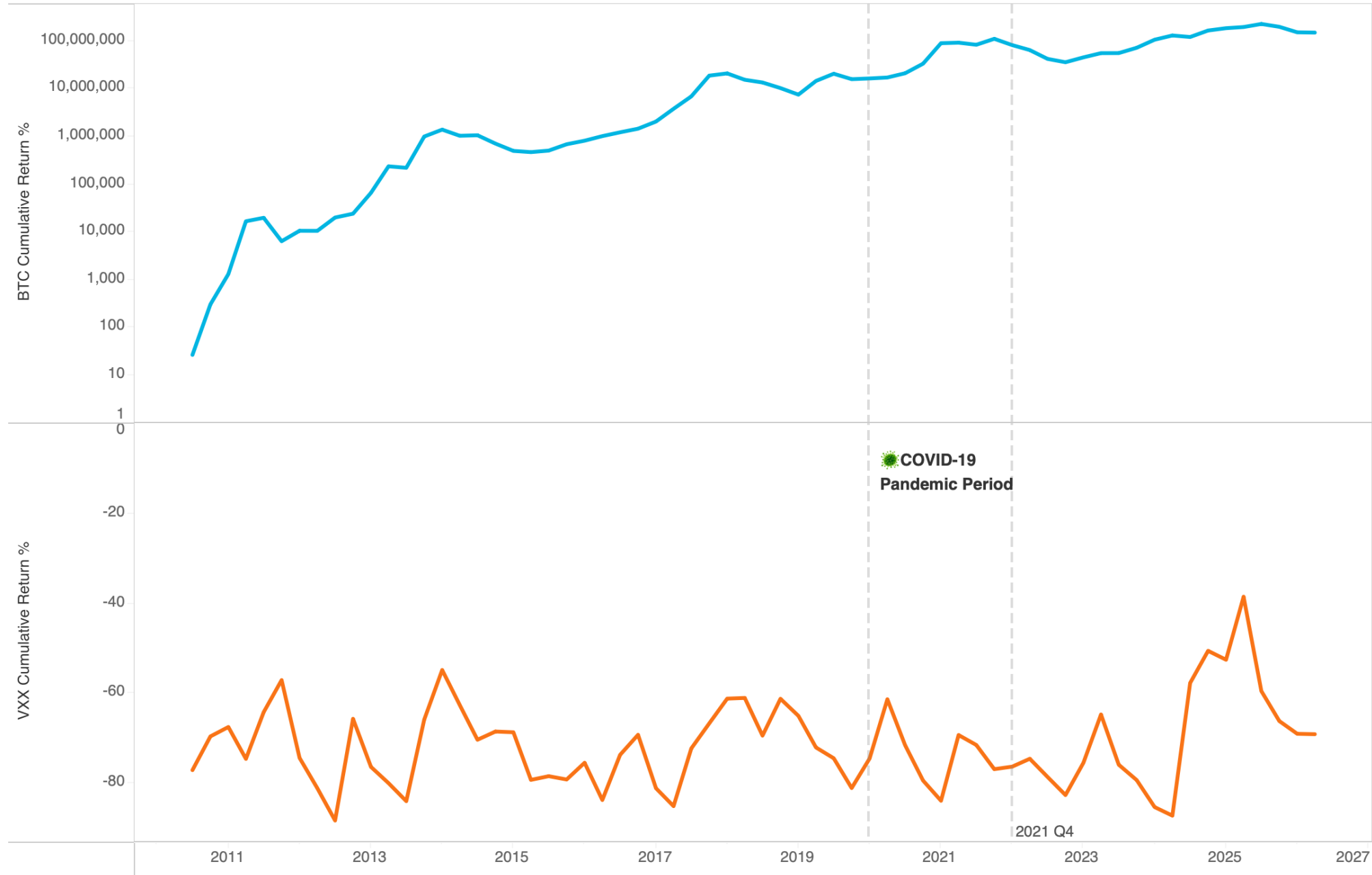
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